Final Conceptual Diagram — Five Candidate Designs

A Comparison Gallery

June 8, 2026

This gallery draws five candidate designs for the closing *Final Conceptual Diagram* of `homological_understanding.tex`, whose subject is $X = \mathbb{P}^1_{\mathbb{C}}$, read at once as the 2-sphere S^2 and as a complex curve. The pipeline to synthesize is

local data → chain complex $(C^\bullet, d^2{=}0)$ → cohomology H^n → $0{\to}A{\to}B{\to}C{\to}0$ → LES $\xrightarrow{\delta}$ global class

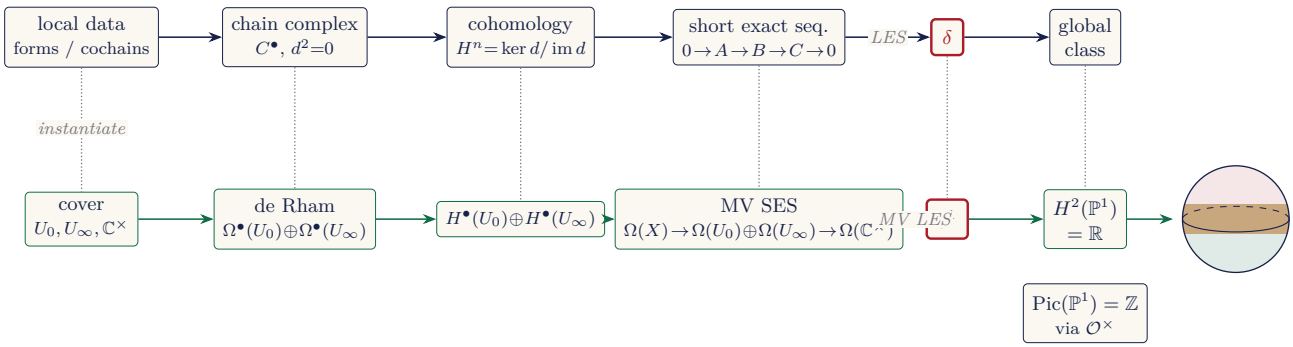
with the Mayer–Vietoris realization

$$0 \to \Omega^\bullet(\mathbb{P}^1) \to \Omega^\bullet(U_0) \oplus \Omega^\bullet(U_\infty) \to \Omega^\bullet(\mathbb{C}^\times) \to 0$$

$(U_0 \cap U_\infty = \mathbb{C}^\times \simeq S^1,\ \delta^1 \colon H^1(\mathbb{C}^\times) \xrightarrow{\cong} H^2(\mathbb{P}^1),\ \mathrm{Pic}(\mathbb{P}^1) = \mathbb{Z})$.

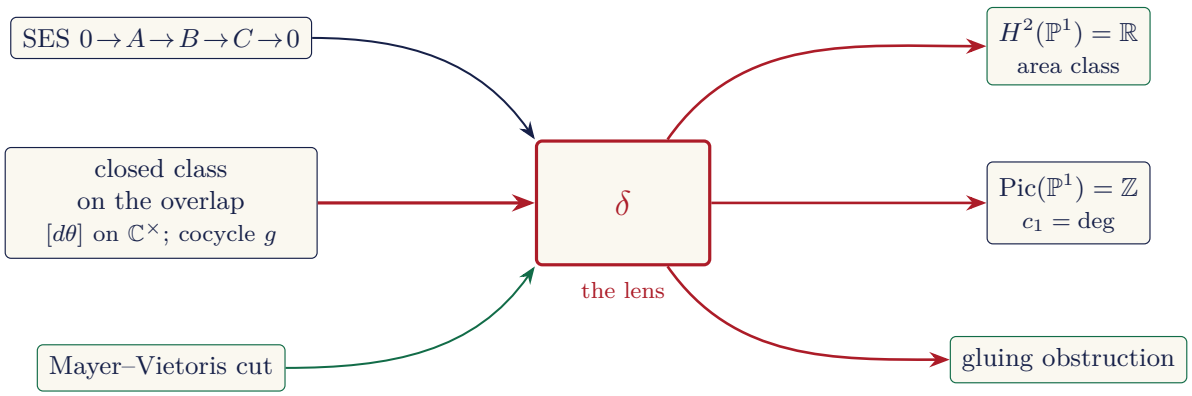
abstract construction geometry / Mayer–Vietoris δ / obstruction

Option 1: Two-track "abstraction ↔ instance" ladder



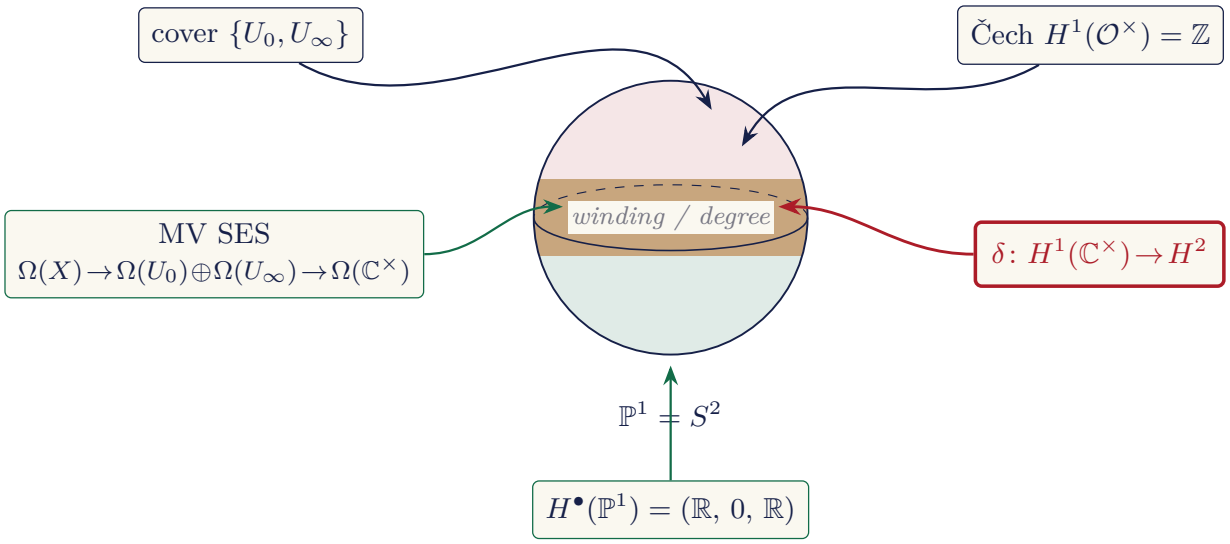
***Best for**: showing the document's thesis — one abstract machine, instantiated on $\mathbb{P}^1$ — as the literal structure; single left-to-right reading, δ as the shared climax, geometric payoff at the end.* ***Trade-off**: the widest design (needs full-width scaling).*

Option 2: The "δ-funnel" (the connecting map as the lens)



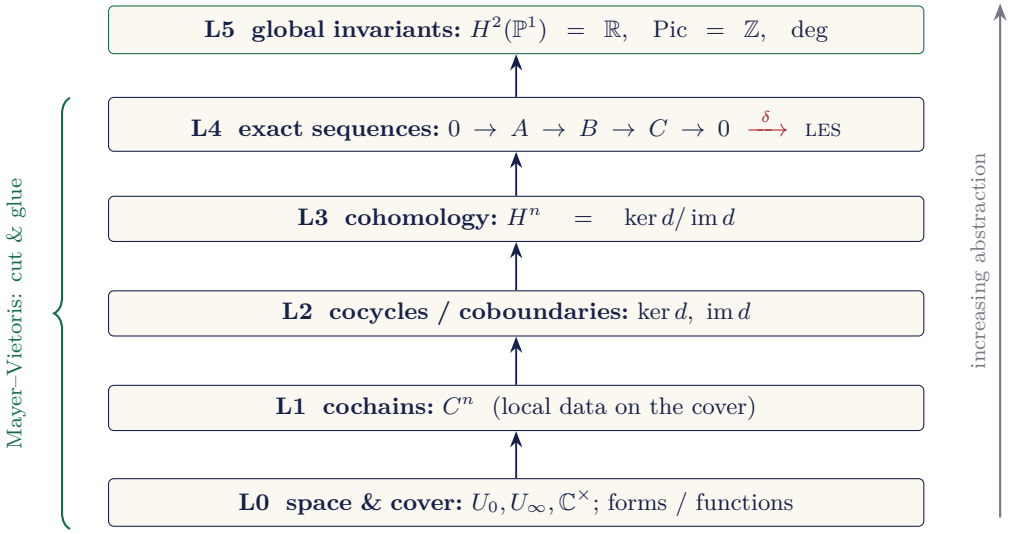
***Best for**: dramatizing the title* Exactness as a Lens *— one construction (δ) drives every result in the document.* ***Trade-off**: de-emphasizes the upstream pipeline (chain complex → cohomology).*

Option 3: Geometry-anchored hub



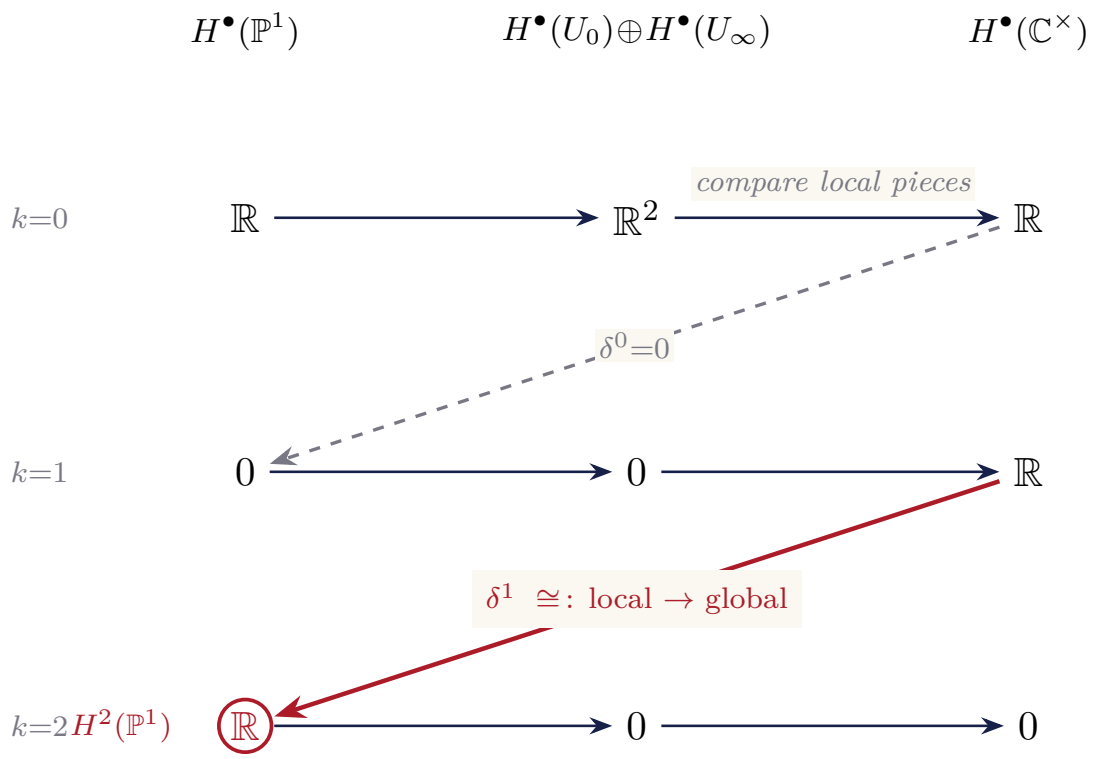
***Best for**: maximal fusion of algebra and the document's geometry — each tool visibly* reads off *a feature of the sphere.* ***Trade-off**: a hub layout (rather than a pipeline); only justified because the hub is a concrete object and every arrow carries a stated "reads-off" meaning.*

Option 4: Vertical "abstraction tower"



***Best for**: a clean, page-friendly ladder of abstraction read bottom-to-top, ending in the concrete invariants.* ***Trade-off**: the "cut and glue" Mayer–Vietoris story survives only as a side annotation.*

Option 5: The zigzag as the literal spine



***Best for**: making the connecting map's zigzag the visible backbone, rhyming with the document's staircase figure.*
***Trade-off**: overlaps heavily with an existing figure, so it adds the least* new *information.*

Comparison

Option	Idea	Reading	Best for	Trade-off
1. Ladder	abstract rail ‖ $\mathbb{P}^1$ rail, joined by rungs	left → right, two rows	the thesis "machine, then instance"; ends in geometry	widest design
2. Funnel	δ as a central lens; inputs in, results out	converge → fan out	the title *Exactness as a Lens*	hides the upstream pipeline
3. Hub	sphere at center; tools read off its features	radial, inward arrows	fusing algebra with the geometry	not a pipeline (hub layout)
4. Tower	strata from concrete to abstract	bottom → top	the ladder of abstraction; compact	MV only a side note
5. Spine	the LES staircase as backbone	zigzag	the snake as the visible spine	duplicates an existing figure

Recommendation. *Option 1 (the two-track ladder)* is the strongest *final synthesis*: it makes the document's central pedagogy — an abstract machine made concrete on $\mathbb{P}^1$ — the literal shape of the picture, reads in one direction, places δ as the shared climax of both rails, and terminates in the geometric payoff ($H^2(\mathbb{P}^1) = \mathbb{R}$, Pic $= \mathbb{Z}$, the sphere). If a punchier, more thesis-driven image is preferred, graft *Option 2's* right-hand fan-out onto Option 1's lower-right end so the lens and its outputs are explicit.